

Pearson Edexcel Award

Tuesday 27 June 2023

Afternoon (Time: 3 hours)

Paper
reference

9811/01

Advanced Extension Award Mathematics

Insert for questions 5, 6 and 7

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5.

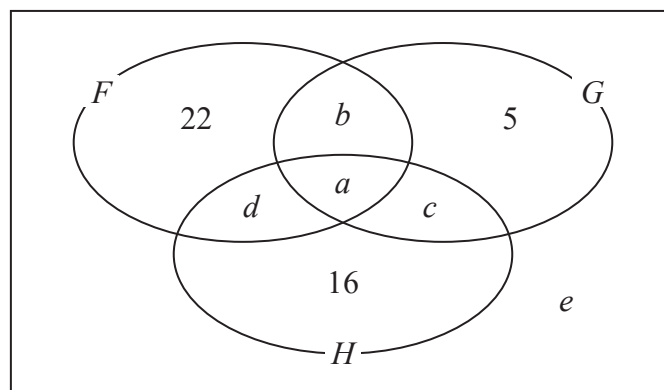


Figure 2

Figure 2 shows a partially completed Venn diagram of sports that a year group of students enjoy, where a , b , c , d and e are non-negative integers.

The diagram shows how many students enjoy a combination of football (F), golf (G) and hockey (H) or none of these sports.

There are n students in the year group.

It is known that

- $P(F) = \frac{3}{7}$
- $P(H | G) = \frac{1}{3}$
- F is independent of $H \cap G$

(a) Show that $P(F \cap H \cap G) = \frac{1}{7} P(G)$ (3)

(b) Prove that if two events X and Y are independent, then X' and Y are also independent. (3)

(c) Hence find the value k such that $P(F' \cap H \cap G) = kP(G)$ (2)

(d) Show that $c = \frac{4}{3}a$ (2)

Given further that $P(F | H) = \frac{1}{5}$

(e) find an expression for d in terms of a , and hence deduce the maximum possible value of a . (3)

(f) Determine the possible values of n . (6)
(+S2)

(Total for Question 5 is 21 marks)

6.

In this question you may assume the following formulae for the volume and curved surface area of a cone of base radius r and height h and of a sphere of radius r .

Cone: volume $V = \frac{1}{3}\pi r^2 h$ and curved surface area $S = \pi r \sqrt{h^2 + r^2}$

Sphere: volume $V = \frac{4\pi}{3}r^3$ and curved surface area $S = 4\pi r^2$

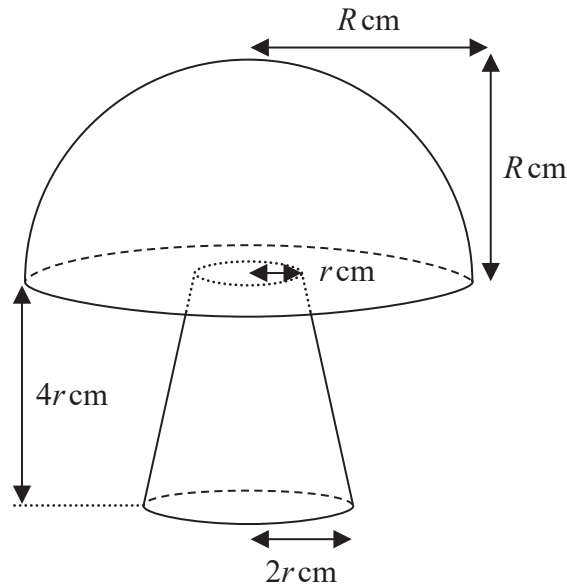


Figure 3

Figure 3 shows the design for a garden ornament.

The ornament is made of a hemisphere on top of a truncated cone.

The truncated cone has base radius $2r$ cm, top radius r cm and height $4r$ cm.

The hemisphere has radius R cm.

Given that the volume of the ornament is $2100\pi \text{ cm}^3$

(a) show that

$$R^3 = 3150 - 14r^3 \quad (5)$$

(b) Find an expression involving $\frac{dR}{dr}$ in terms of r and/or R .

(2)

The base of the truncated cone of the ornament is fixed to the ground.

(c) Show that the visible surface area of the ornament, $A \text{ cm}^2$, is given by

$$A = (3\sqrt{17} - 1)\pi r^2 + 3\pi R^2 \quad (5)$$



Question 6 continued

(d) Hence show that

$$\frac{dA}{dr} = \gamma\pi r - \frac{\delta\pi r^2}{R}$$

where γ and δ are real numbers to be determined.

(2)

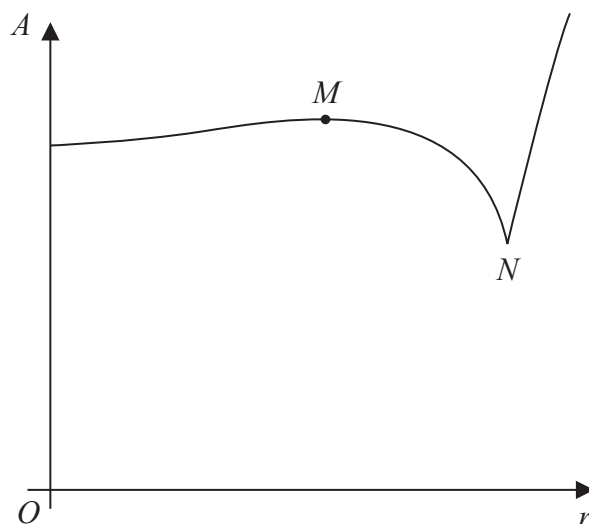


Figure 4

Figure 4 shows a sketch of A against r , for $r \geq 0$

There is a local minimum at $r = 0$ and a local maximum at the point M . The overall minimum point is at the point N , where the gradient of the curve is undefined.

(e) (i) Determine the r coordinate of the point N .

(ii) Explain why, for the ornament, r must be less than this value.

(2)

(f) Show that the r coordinate of the point M is

$$\sqrt[3]{\frac{p(3\sqrt{17}-1)^3}{3q^2 + (3\sqrt{17}-1)^3}}$$

where p and q are integers to be determined.

(5)

(+S2)

(Total for Question 6 is 23 marks)

7. A sequence of non-zero real numbers $a_1, a_2, a_3, \dots$ is defined by

$$a_{n+1} = p + \frac{q}{a_n} \quad n \in \mathbb{N}$$

where p and q are real numbers with $q \neq 0$

It is known that

- one of the terms of this sequence is a
- the sequence is periodic

(a) Determine an equation for q , in terms of p and a , such that the sequence is constant (of period/order one).

(2)

(b) Determine the value of p that is necessary for the sequence to be of period/order 2.

(3)

(c) Give an example of a sequence that satisfies the condition in part (b), but is **not** of period/order 2.

(2)

(d) Determine an equation for q , in terms of p only, such that the sequence has period/order 4.

(7)

(+S1)

(Total for Question 7 is 15 marks)



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